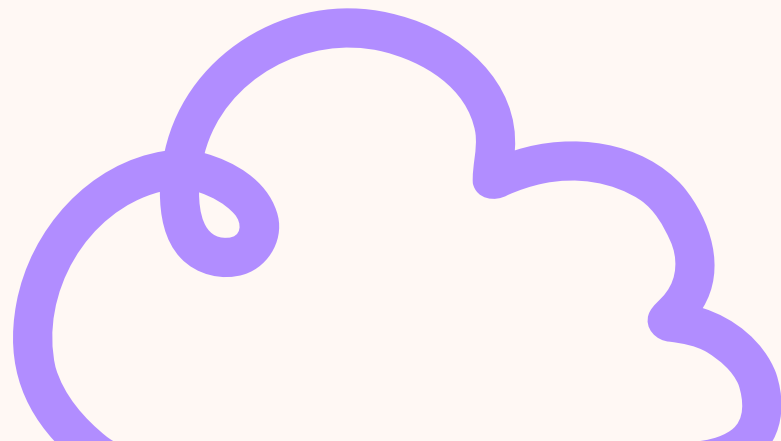
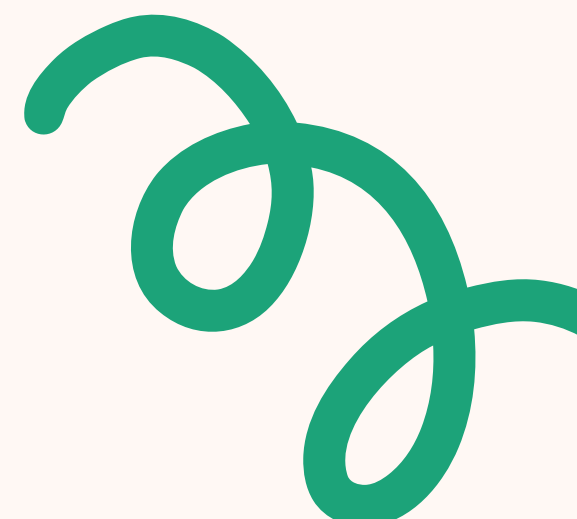
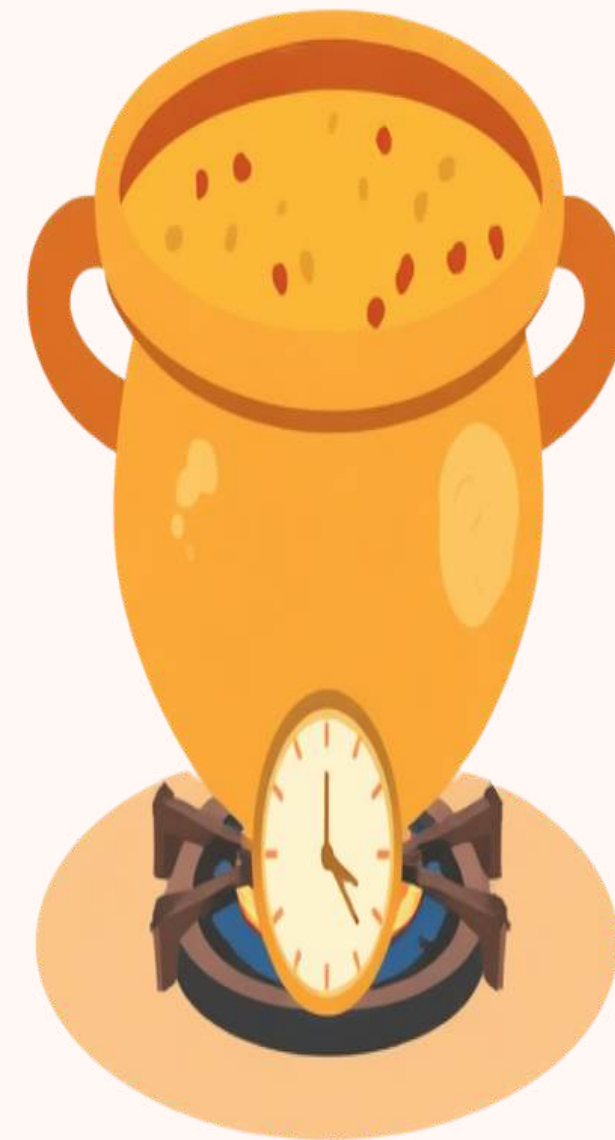


CATALYSTS — THE SPEED BOOSTERS OF SCIENCE



Fast or Slow? The Mystery Reaction!

Why does tea boil fast but kheer takes hours?
What makes some reactions quick and others slow?

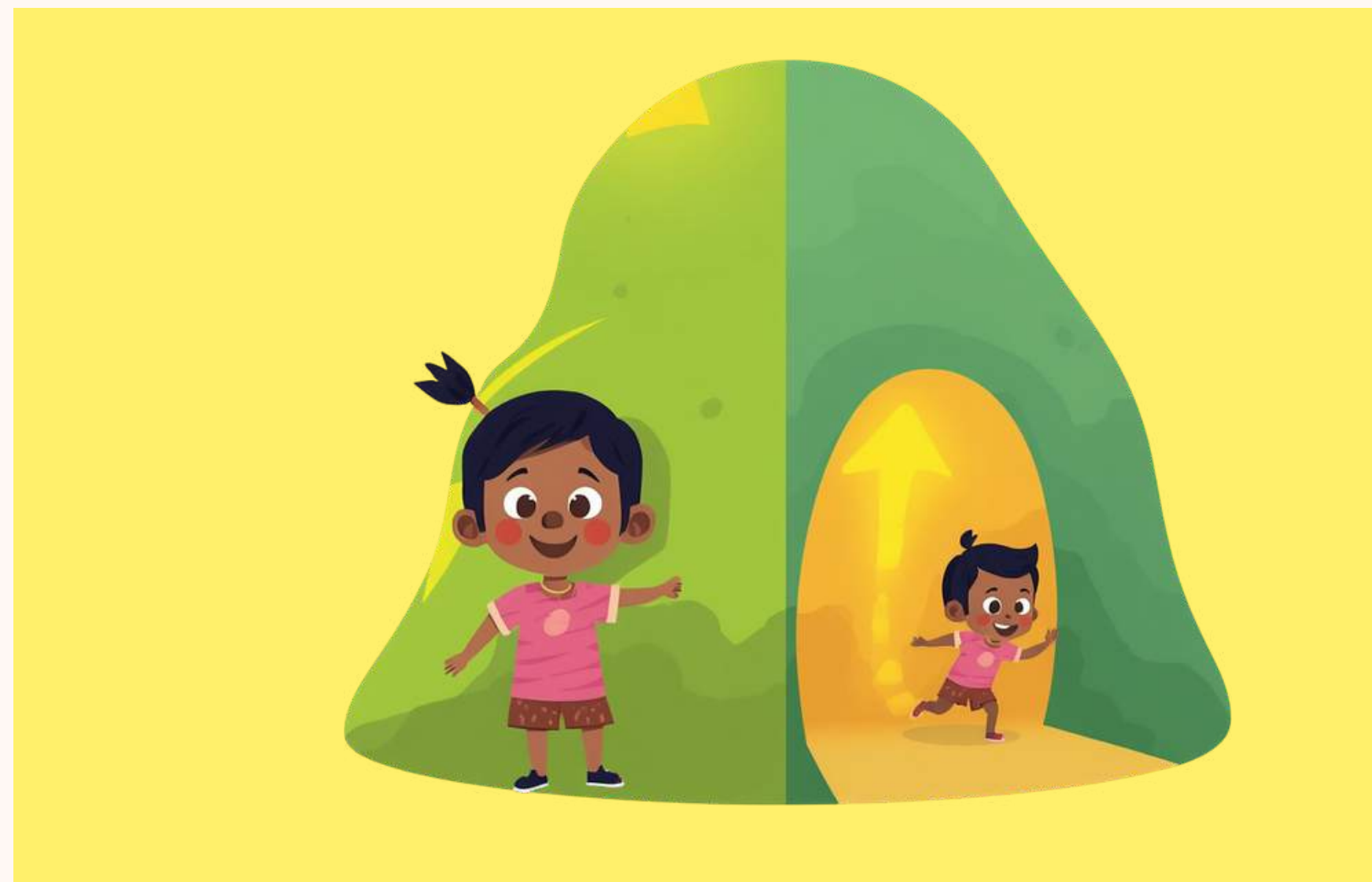


Climbing the Hill or Taking the Tunnel?

Reactions need energy to start — like climbing a big hill! 🏔️

A catalyst helps find a shortcut through the tunnel — same destination, less effort!

⚡ Less energy needed = faster reaction!



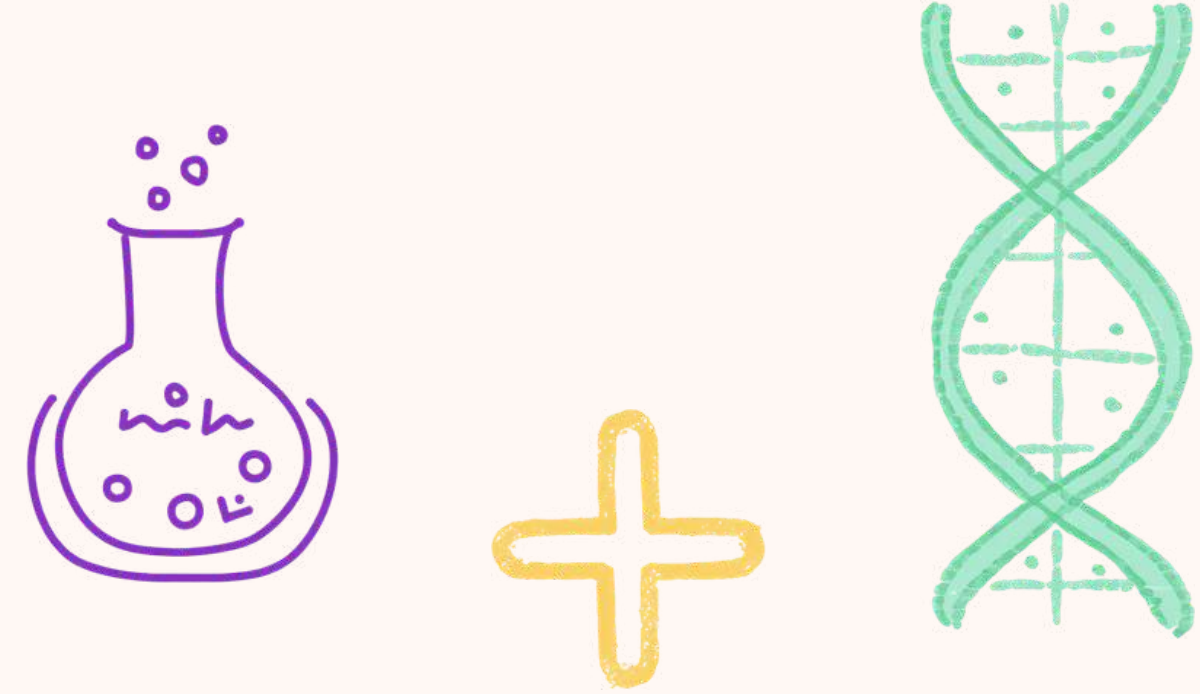
What is a Catalyst?

A catalyst speeds up a reaction without being used up or changed! Reactants go in → Products come out — and the catalyst stays exactly the same. ✨

Think of it as a helpful friend who makes things happen faster!



Enzymes vs Chemical Catalysts



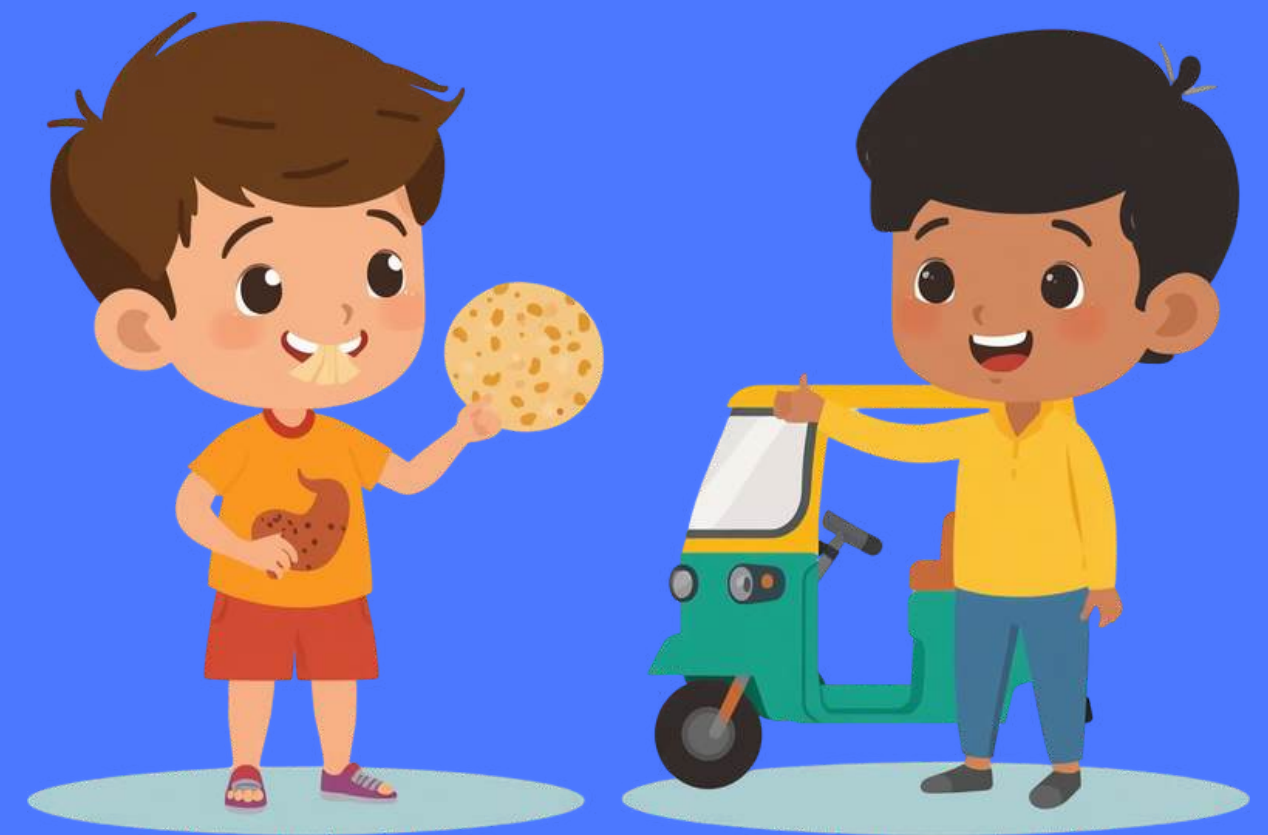
Enzymes (In Our Body)

Saliva breaks down chapati as you chew!

Chemical Catalysts (Labs & Cars)

Catalytic converter cleans auto-rickshaw exhaust!

Both speed up reactions but work in different places.





Everyday Catalysts Around Us



💧 Pressure Cooker

Steam speeds up cooking!

🥣 Idli-Dosa Batter

Yeast ferments the batter overnight.

💡 Diwali Sparkler

Burns fast vs. iron gate rusting slow.

🚗 Catalytic Converter

Cleans pollution from auto-rickshaws!





Safe Science Demo: Elephant Toothpaste!



Watch how a catalyst (yeast) speeds up foam creation using hydrogen peroxide and dish soap!

⚠ Safety Tips:

- Adult supervision needed at all times.
- Use only safe, approved materials.
- Keep away from eyes and mouth.



Let's Find the Catalyst!

Look at these 4 scenes — which ones have a catalyst helping the reaction?

- 🍵 Tea boiling on flame
- 🥫 Idli batter fermenting
- 🔩 Rusting iron gate
- 💫 Sparkler burning
- 👉 Which ones use a catalyst? Think carefully!



Recap: Why Are Catalysts Amazing?

- ⚡ Speed up reactions
- ✓ Not used up or changed
- 🌱 Help save energy (less heat, less time)
- 🌍 Found everywhere — kitchen, body & environment



Your Home Explorer Task!

Watch your cooking or garden for catalysts!

Ask an adult to help with simple experiments.

Look for bubbling batter, boiling tea, or a sparkler — catalysts are all around you!





Thank You & Keep Exploring!

Great job today! Keep discovering
science all around you.

